

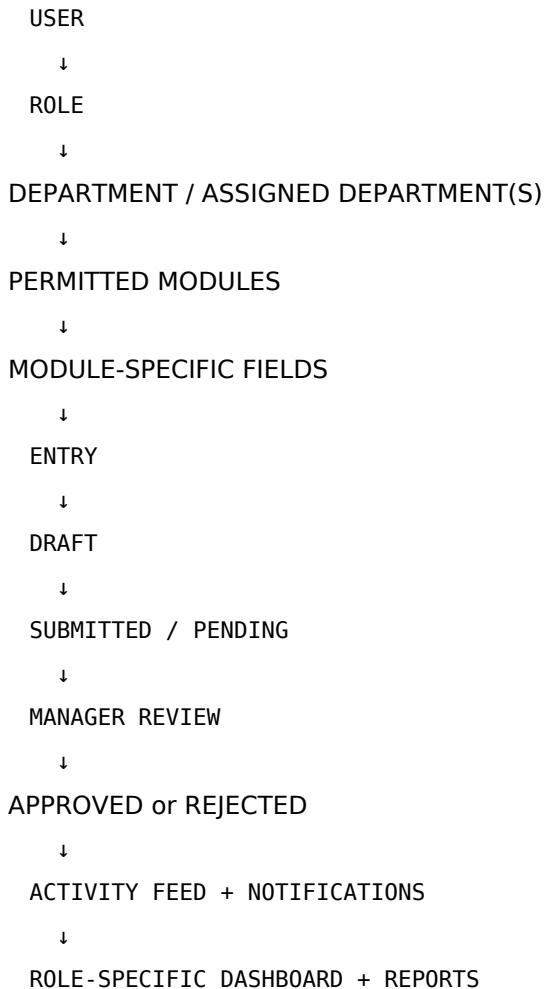
RIVET IMPLEMENTATION BLUEPRINT

Source of Truth: RIVET PRD

Purpose: Reshape the existing RIVET application to match the PRD exactly, without breaking working functionality or inventing unsupported features.

1. CORE APPLICATION HIERARCHY

RIVET must work according to this hierarchy:



Important:

- Access must be determined by the authenticated user's stored role, department assignment, and permissions.
- Never determine role from the email address.
- Never allow a public user to choose their own role.
- Admin controls role assignment, department assignment, and module configuration.
- Existing working authentication/dashboard/navigation should be preserved where possible and reshaped rather than unnecessarily rebuilt.

2. ROLES

EMPLOYEE

Purpose:

- Perform work.
- Create and maintain permitted entries.

- View their own work/history.
- Submit entries for approval.
- Correct rejected entries and resubmit.
- Receive relevant notifications.

Access:

- Only their assigned department's permitted modules.
- No access to other departments' private working data.
- Shared/view-only modules must follow their specific PRD permissions.

MANAGER

Purpose:

- Manage and review work for assigned department(s).
- Review employee submissions.
- Edit entries before approval where permitted.
- Approve or reject entries.
- See team/department summaries.
- Use approval List and Board/Kanban views.

Access:

- Assigned department(s).
- Employees belonging to those departments.
- Department-level data and approvals.
- No unrelated department data unless explicitly allowed by role/assignment.

ADMIN

Purpose:

- Organization-wide system administration.

Access:

- All departments.
- All records.
- Users.
- Departments.
- Module configuration.
- Cross-department dashboards.
- Reports/exports.
- Audit log.
- System health.

Important:

- A Manager may be assigned to multiple departments.
- Admin assignment controls must support this.

3. DEPARTMENT → MODULE MAP

RESEARCH DEPARTMENT

1. Daily Activity Log

DATA MANAGEMENT DEPARTMENT

- 1. Weekly Meeting Report**
- 2. Daily Meeting Attendance Log**
- 3. 52 Week Webinar Series Planner**
- 4. Branch-Wise Meeting Data**

BRAND & MARKETING DEPARTMENT

Detailed PRD sections:

- 1. Zoom / Webinar Link Management**
- 2. Data Management Log**
- 3. Call Log**
- 4. Branding Activities Log**
- 5. Social Media Management Log**
- 6. Campaign hierarchy / parent-child structure**

PRD NOTE:

The PRD text contains a wording inconsistency where it refers to "four modules" but separately specifies the detailed module areas above. Do not remove any explicitly documented detailed section. Keep all detailed definitions present and flag this count mismatch for final PRD clarification before production if necessary.

4. MODULE RULE

The Modules page must NEVER show a generic list of every module to every user.

Instead:

Employee

- assigned department
- modules permitted for that department
- only actions allowed for Employee

Manager

- assigned department(s)
- modules/data permitted for those departments
- manager actions

Admin

- all departments
- all configured/enabled modules
- administrative configuration and organization-wide access

Module access and ACTION permissions are separate.

Example:

A module may be visible to an Employee but only viewable, while Manager/Admin can create or edit it.

5. RESEARCH MODULE

MODULE: DAILY ACTIVITY LOG

Fields:

- Date
- Day of Week (auto-computed)
- Activity Description
- Status
- Manager Remarks

Rules:

- Prevent duplicate entries for the same required date.
- Employee can edit their own entry until manager approval.
- Submitted entry becomes pending.
- Rejected entry can be corrected and resubmitted.
- Manager remarks must be shown when relevant.

Employee flow:

- Research Employee
 - Research Dashboard
 - Daily Activity Log
 - Create Entry
 - Save Draft
 - Submit
 - Manager Review
 - Approved / Rejected

6. DATA MANAGEMENT MODULES

MODULE 1: WEEKLY MEETING REPORT

Fields:

- Week Ending Date (Friday)
- Physical Meetings
- Online Meetings
- Calls
- Service Calls
- Service Meetings

Rules:

- Monthly and cumulative totals calculated automatically.
- Manager can edit before approval.

MODULE 2: DAILY MEETING ATTENDANCE LOG

Fields:

- Date
- Physical Meetings Attended
- Zoom Meetings Attended

- Beyond Sessions Attended

Rules:

- Manager must be able to see daily/monthly totals per employee and for the team.

MODULE 3: 52 WEEK WEBINAR SERIES PLANNER

Fields:

- Series Number (auto-assigned)
- Scheduled Date
- Topic Title
- Speaker Name
- Speaker Designation / Experience
- Event Mode
- Status
- Notes

Event Mode:

- Online
- Physical
- Hybrid

Status:

- Planned
- Confirmed
- Completed
- Cancelled

Rules:

- Shared department-level calendar.
- Manager/Admin can create and edit.
- Employee can view.
- Employee can submit future-topic suggestions.
- This is NOT treated like a personal daily employee log.

MODULE 4: BRANCH-WISE MEETING DATA

Fields:

- Month
- Branch Name
- Meeting Type
- Count

Meeting Type:

- Physical
- Online
- Call

Purpose:

- Regional/branch-level reporting.

7. BRAND & MARKETING MODULES

MODULE 1: ZOOM / WEBINAR LINK MANAGEMENT

Fields:

- Date
- Meeting Title / Purpose
- Host / Organiser
- Scheduled Date
- Scheduled Time
- Duration
- Zoom Link / Meeting ID
- Sent To
- Status
- Notes
- Attendee Count

MODULE 2: DATA MANAGEMENT LOG

Fields:

- Date
- Task / Data Set Description
- Type
- Source / System
- Action Taken
- Status
- Records Handled (optional)
- Verified By (optional)
- Remarks

MODULE 3: CALL LOG

Fields:

- Date
- Company / Contact Name
- Client Type
- Phone / Email
- Call Time
- Duration
- Purpose of Call
- Outcome / Result
- Follow-up Required?
- Notes

MODULE 4: BRANDING ACTIVITIES LOG

Fields:

- Date
- Activity / Task Title
- Activity Type
- Platform / Medium
- Status
- Time Spent
- Deliverable / Output Description
- Remarks

Activity Type:

- Video

- Image
- Copy
- Campaign

Status:

- In Progress
- Done
- On Hold

MODULE 5: SOCIAL MEDIA MANAGEMENT LOG

Fields:

- Date
- Post / Content Title
- Platform
- Content Type
- Scheduled Time
- Status
- Likes (optional)
- Comments (optional)
- Shares / Reach (optional)
- Remarks

Platform:

- Instagram
- LinkedIn
- YouTube
- WhatsApp
- Other

Content Type:

- Image
- Video
- Reel
- Story
- Text

Status:

- Scheduled
- Posted
- Delayed
- Cancelled

8. CAMPAIGN HIERARCHY

Campaign is a parent entity.

CAMPAIGN

- ├ Branding Activity entries
- ├ Social Media entries
- └ Other linked Brand & Marketing child entries

Campaign fields:

- Campaign Name

- Start Date
- End Date
- Platform / Medium
- Status
- Owner
- Description

Rules:

- Child entries may optionally link to an existing campaign.
- Campaign detail groups linked child records.
- Campaign-level aggregate statistics must be calculated from linked records according to the PRD.

9. ENTRY LIFECYCLE

DRAFT

→ Employee can work on the entry.

SUBMITTED / PENDING

- Entry goes to Manager.
- Employee cannot edit while pending.

APPROVED

→ Entry is approved/final.

REJECTED

- Manager provides remarks.
- Employee receives notification.
- Employee corrects the entry.
- Employee resubmits.
- Entry returns to Pending.

MANAGER EDIT:

- Manager may edit an entry before approval where permitted.
- Manager changes must be captured in the audit history.

No entry should be permanently deleted where the PRD requires historical retention.

10. APPROVAL MANAGEMENT

ROUTE:

/app/approvals

MANAGER:

- See submissions for assigned department(s).
- Default List View.
- Optional Board/Kanban View.
- Filter by relevant department/module/status/date.
- Open an entry for full review.
- Approve.
- Reject.
- Rejection requires remarks.
- Bulk approve/reject where supported by the PRD.

ADMIN:

- Can view approval board/data across departments.
- Can switch the department being viewed.

BOARD STATUS:

- Pending
- Approved
- Rejected

APPROVAL DETAIL:

- Show complete entry data.
- Show employee.
- Show department.
- Show module.
- Show status.
- Show submission information.
- Show manager remarks where applicable.
- Show allowed actions for the current role.

11. APPLICATION ROUTES / SCREENS

PUBLIC ROUTES

/

Public marketing/home page.

SIGN UP

/sign-up

- Account creation form.
- No public role selection.
- Department selection may exist in the prototype.
- Successful signup shows confirmation.
- User must then sign in.

SIGN IN

/sign-in

- Email
- Password
- Show/hide password.
- Authentication determines the user's stored role and department.
 - Successful authentication → correct role-specific dashboard.
 - Invalid credentials → remain on sign-in with clear error.
- No direct dashboard access without valid authentication.

Production authentication later must support the PRD authentication requirements, including proper account/session handling and Google authentication.

AUTHENTICATED APP

/app/dashboard

Role-specific dashboard.

`/app/work`

My Work / personal accessible work.

`/app/modules`

Accessible modules for the current user.

`/app/modules/:moduleKey`

Module detail.

`/app/modules/:moduleKey/new`

Create entry where permitted.

`/app/modules/:moduleKey/entry/:entryId`

Entry detail.

`/app/approvals`

Manager/Admin approval management.

`/app/approvals/:entryId`

Approval detail.

`/app/reports`

Role-scoped reporting and exports.

`/app/notifications`

Notifications/activity events.

`/app/settings`

Only supported account/application settings.

`/app/admin/users`

Admin-only user management.

`/app/admin/departments`

Admin-only department management.

`/app/admin/modules`

Admin-only module configuration.

`/app/admin/audit`

Admin-only audit log.

IMPORTANT:

- Keep existing TanStack Router architecture.
- Do not create duplicate route systems.
- Protect all `/app/*` routes.
- Unauthorized role access must be blocked.
- Directly entering a protected URL must still enforce RBAC.

12. BACK ARROW NAVIGATION

Use a simple arrow only:

←

Do NOT write:

- Back to Modules
- Back to My Work
- Back to Approvals

Show the arrow on appropriate nested/detail pages.

Examples:

Modules → Module Detail

Module Detail → New Entry

My Work → Entry Detail

Approvals → Approval Detail

Rules:

- Arrow returns to the logical previous/parent page.
- Browser Back/Forward must continue to work normally.
- No redirect loops.
- No forced page reloads.
- Do not create multiple competing navigation systems.

13. EMPLOYEE DASHBOARD

The Employee Dashboard is PERSONAL + DEPARTMENT-AWARE.

Show:

- Activity Feed at the top.
- Recent submissions.
- Approval/rejection activity.
- Relevant comments/mentions.
- Reminder events.
- Personal work/log history.
- Date/module/status filtering.
- Monthly summary.
- Trend information.
- Upcoming webinar series where relevant.

Module area:

- Only modules available to the employee through their assigned department and permissions.

Example:

Research Employee

→ Research-related access

→ Daily Activity Log

→ Personal activity/history

Data Management Employee

→ Data Management modules

→ Personal activity/history

→ Shared webinar planner view where permitted

Brand & Marketing Employee

→ Brand & Marketing modules permitted to the employee

→ Personal activity/history

Every activity item should open the relevant record/detail page where applicable.

14. MANAGER DASHBOARD

Manager Dashboard is TEAM + DEPARTMENT focused.

Show:

- Activity Feed.
- Team activity.
- Employees in assigned department(s).
- Submission status.
- Pending approval queue.
- Department monthly summary.
- Meeting totals.
- Team reporting.
- Relevant module summaries.

TEAM GRID:

- Rows = Team members
- Columns = Days of week
- Cells = Approved / Pending / No Entry / Weekend-Holiday

Interaction:

- Clicking a populated cell opens its entry.
- Empty cells may open permitted pre-filled entry creation where supported.

Manager must only see team/department information permitted by their assignments.

15. ADMIN DASHBOARD

Admin Dashboard is ORGANIZATION-WIDE.

Show:

- Cross-department summary.
- Active users.
- Last login.
- Entry counts.
- System health.
- Notification delivery status.
- Failed export information.
- Relevant cross-department activity.

Admin also has access to:

- Users
- Departments
- Modules
- Audit
- Organization-wide reporting

16. ACTIVITY FEED

Location:

Top of the home/dashboard experience.

EMPLOYEE:

- Own relevant activity.

MANAGER:

- Activity relevant to assigned departments.

ADMIN:

- Cross-department activity.

Feed requirements:

- Clickable activity items.
- Group by date.
- Refresh every 60 seconds.
- Show an indication/banner when new activity arrives.

17. NOTIFICATIONS

Events:

Employee:

- Missing daily submission.
- Entry approved.
- Entry rejected.

Manager:

- New submission.
- Multiple pending entries.

Other:

- New department member notification where applicable.

Notification delivery:

- In-app.
- Email where specified by event/configuration.

Reminder configuration:

- Default daily reminder: 6 PM.
- Manager digest: 9 AM when pending queue is non-empty.

Notifications must connect to the actual underlying entry/event.

18. GLOBAL SEARCH

Persistent search in top navigation.

ADMIN:

- All departments/modules/data.

MANAGER:

- Assigned department(s).

EMPLOYEE:

- Their own accessible data.

Search should support relevant indexed fields such as:

- Employee name
- Module
- Date
- Activity description

- Task title
- Call purpose
- Other module-specific searchable fields

Support date range filtering where specified.

Do not create a fake search bar that does nothing.

19. REPORTS & EXPORTS

Required report types:

1. Monthly Department Report

2. Individual Employee Report

3. Weekly Meeting Summary

4. 52 Week Calendar

5. Full Data Export (Admin)

Formats:

- Excel (.xlsx)
- PDF where specified by the PRD

Scope must follow role permissions.

Do not show export buttons unless the corresponding export function is implemented.

20. ADMIN CONFIGURATION

USERS

- Create.
- Edit.
- Deactivate.
- Assign role.
- Assign department(s).
- View account information.

DEPARTMENTS

- Create.
- Configure.
- Assign modules.

MODULES

- Create/configure.
- Enable/disable.
- Assign to departments.

New departments should be addable without redesigning the application core.

21. AUDIT LOG

Admin-only.

Record:

- Who performed the action.

- What action occurred.
- Which record was affected.
- Timestamp.
- Changed fields/details where applicable.

Tracked actions:

- Create
- Edit
- Approve
- Reject

Retention:

- At least the retention period defined by the PRD.

22. DATA / SECURITY RULES

Production user data must store:

- User identity
- Role
- Department assignment(s)
- Account status
- Authentication information
- Relevant permissions

Never use:

- Email naming conventions to determine role.
- Frontend-only role checks as the real security boundary.
- Public role selection as authority.

Final architecture:

Frontend

→ React

Backend

→ FastAPI

Database

→ PostgreSQL / Neon

Authentication

→ JWT + Google authentication as specified by PRD

Background processing / notifications

→ Redis-based queue/scheduling architecture as defined by PRD

Backend must enforce:

- Authentication.
- Authorization.
- Department access.
- Module access.
- Entry ownership.
- Approval permissions.
- Data validation.

Frontend guards improve UX, but backend authorization is the real security layer.

23. UI / UX RULES

- Keep the existing RIVET professional visual identity.
- Do not redesign unrelated pages unnecessarily.
- Do not add random features.
- Do not add fake KPI cards.
- Do not add dead buttons.
- Do not add decorative UI that has no product purpose.
- Use clear hierarchy and consistent spacing.
- Keep navigation predictable.
- Use only actions that are actually supported.
- Every button must either perform its intended function or not be shown.
- Preserve responsive desktop/tablet/mobile behavior.

24. IMPLEMENTATION ORDER

PHASE 1 - FOUNDATION

- Role model.
- Department model.
- Module access.
- Permission model.
- Navigation.
- Protected routes.

PHASE 2 - RESEARCH

- Daily Activity Log.
- Exact fields.
- Validation.
- Entry lifecycle.

PHASE 3 - DATA MANAGEMENT

- Weekly Meeting Report.
- Daily Meeting Attendance Log.
- 52 Week Webinar Series Planner.
- Branch-Wise Meeting Data.
- Required calculations.

PHASE 4 - BRAND & MARKETING

- Zoom/Webinar Link Management.
- Data Management Log.
- Call Log.
- Branding Activities Log.
- Social Media Management Log.
- Campaign hierarchy.

PHASE 5 - APPROVAL SYSTEM

- List view.
- Board/Kanban view.
- Approve.
- Reject + mandatory remarks.
- Resubmission.

- Manager edit-before-approval.
- Audit tracking.

PHASE 6 - DASHBOARDS

- Employee.
- Manager.
- Admin.
- Activity Feed.
- Real data instead of demo-only information.

PHASE 7 - NOTIFICATIONS

- In-app events.
- Reminder logic.
- Manager digest.
- Email integration when backend is implemented.

PHASE 8 - SEARCH / REPORTS / EXPORTS

- Global Search.
- Required report types.
- Excel/PDF exports.

PHASE 9 - ADMIN

- User management.
- Department management.
- Module management.
- Audit log.
- System health.

PHASE 10 - PRODUCTION AUTH / BACKEND

- FastAPI.
- PostgreSQL/Neon.
- JWT.
- Google authentication.
- Real RBAC.
- Real persistent user/department/module data.
- Replace temporary demo authentication safely.

25. NON-NEGOTIABLE IMPLEMENTATION RULE

Do NOT rebuild everything in one uncontrolled change.

Implement one phase at a time.

After every phase:

- Type-check.
- Build.
- Verify routes.
- Verify role access.
- Verify department access.
- Verify module visibility.
- Verify important workflows.
- Confirm existing functionality still works.

Never solve a problem by:

- deleting existing working features,
- bypassing RBAC,
- adding redirect hacks,
- duplicating state/route systems,
- replacing local work with remote/generated code,
- or inventing requirements not present in the PRD.

FINAL PRODUCT PRINCIPLE:

Every screen, module, field, action, dashboard item, approval action, notification, report and data view must be controlled by:

ROLE

+ DEPARTMENT

+ MODULE

+ PERMISSION

+ ENTRY STATE

This is the master structure that RIVET must follow.